

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	534257
Award Program	New Innovator Award in Food and Agriculture Research
Project Title	Innovation at the nexus of computational, membrane and plant biology
Grant Start Date	10/01/2017
Grant End Date	09/30/2021
Total Grant Budget	\$600,000
Total FFAR Budget	\$300,000
Matching Funder(s)	University of Illinois
Final Report Date	04/06/2023
Project Director/Principal Investigator Information	
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Grantee Organization Information	
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ASO Title	Vice president/chief financial officer and comptroller of the University of Illinois System
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General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.

Champaign, IL, IL-13th Congressional district.

- 1.2. How many new jobs were created because of this grant funding?

Graduate Student – 3; Postdoctoral Fellow -1;

- 1.3. How many jobs were maintained because of this grant funding?

Graduate Student – 3; Postdoctoral Fellow -1;

- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

No.

- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

No.

2. Scientific Report



- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

One hundred billion kilograms of nitrogenous fertilizers are used worldwide every year as modern agriculture practices entail applying large amounts of inorganic nitrogenous fertilizer to maximize crop yield. However, plants only take up about 40% of the applied nitrogen from fertilizers, and the rest is lost to the environment, resulting in extensive air and water pollution. Environmental nitrogen is supplied to plants mainly in the form of nitrate ions via membrane transporter proteins from the PTR family of major facilitator superfamily (MFS) of transporters. These transporters undergo large conformational changes when transferring nitrate from soil to root. Despite the functional importance of the plant nutrient transport, only one crystal structure of the key nitrate transporter in *Arabidopsis thaliana* is available. A single crystal structure of plant nitrate transporter cannot provide the mechanistic basis of nitrate sensing, uptake and regulation; consequently, we do not know how to engineer transporters with enhanced N-use efficiency. Therefore, the objectives of this project to obtain mechanistic understanding of plant nitrogen uptake process and its regulation using novel computational biophysics approaches. To the best of PI's knowledge, there are no reported molecular modeling studies of the plant nutrient transport in literature. The proposed project will not only utilize (for the first time) modern molecular simulation and machine learning approaches to address a key agricultural challenge but also provide a unique platform to demonstrate the need, feasibility and impact of computational biophysics in addressing fundamental challenges in agriculture.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-wordlimit).

This project is well-aligned with the FFAR goal of improving plant efficiency and takes the first step towards understanding the molecular origin of one of the bottlenecks in the complicated puzzle of nitrogen/fertilizer use efficiency. Food production leads to one-third of all greenhouse emissions and approximately half of the environmental impact of producing the loaf of bread is due to the use of ammonium nitrate fertilizer. Therefore, efficient use of



fertilizers is essential for future of sustainable food production. Overall, understanding of biophysical mechanisms of membrane proteins even for model plants are less complete as compared to humans or any other species of high biological importance. Therefore, the design of proteins with a desired function is done primarily using trial-and-error. Computational approaches combined with molecular level biophysical experiments can provide this much-needed information. This proposal demonstrates how we can successfully implement this strategy for a food and agricultural research problem of high importance to society.

2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;
- b) b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and
- c) c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

The findings of this study could be used to make improvement in future investigations and research. In particular, there are three areas that could benefit from the findings.

1. Software’s developed as part of this project would facilitate research on plant proteins. We have developed software’s to
 - (1) Design Optimal experiments to probe the dynamics of membrane proteins based on simulation data. This software could be used to study any plant protein for which molecular dynamics simulation data is available.
 - (2) Perform simulations of proteins to obtain their structure using low resolution SAXS data. Plant protein structures represent only 2% of all reported structures in the protein data bank (pdb.org). Therefore, this SAXS guided enhanced sampling approach could provide much needed tool obtain structural information when only low resolution experimental data is available.
 - (3) Obtain predictions of multiple conformations of the proteins using only



sequence data. We have developed Fingerprint contacts algorithm which can circumvent the need for complicated crystallography experiments by predicting the multiple conformations of the same protein using their sequence.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

We have made significant progress on the specific aims 1, 2 and 3 of the project in this reporting period:

1. Understanding how plant nitrate transporter NRT1.1 performs high-affinity switching when the nitrate concentration is low.
2. Validating conformational ensemble of transporters using DEER spectroscopy or other experimental data. We have extended the methodologies developed in year 1 and 2 to understand why there is a mismatch between the computational and experimental DEER spectroscopy measurements.
3. Identifying mechanisms responsible for the nitrate uptake efficiency in bacteria and higher plants. We have now simulated the nitrate uptake mechanism in soil bacterial transporters, NarK and NarJ.

The approved application does not list important activities of phases for this reporting period. However, we have made significant progress on specific aim 1 (100%), aim 2 (100%) and aim 3 (100%) by September 2021.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

No changes in major goals/specific aims

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions



(both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)

Major Activities: In this reporting period, we have worked on the three specific aims of the proposal.

Specific Aim 1: Understanding the interplay between receptor dimerization and phosphorylation in regulating nitrate transport.

- a. Improving nitrogen use efficiency is critical to enhancing agricultural productivity and to mitigate environmental pollution. To overcome the fluctuations in soil nitrate concentration, plants have evolved an elaborate nitrate transporting mechanism that switches between high and low affinity. In plants, NRT1.1, a root-associated nitrate transporter, switches its affinity upon phosphorylation at Thr101. However, the molecular basis of this unique functional behavior known as dual-affinity switching remains elusive. Crystal structures of the NRT1.1 nitrate transporter have provided evidence for the two competing hypotheses to explain the origin of dual-affinity switching. It is not known how the interplay between transporter phosphorylation and dimerization regulates the affinity switching. To reconcile the different hypotheses, we have performed extensive simulations of nitrate transporter in conjunction with Markov state models to elucidate the molecular origin for a dual-affinity switching mechanism. Simulations of monomeric transporter reveal that phosphorylation stabilizes the outward-facing state and accelerates dynamical transitions for facilitating transport. On the other hand, phosphorylation of the transporter dimer decouples dynamic motions of dimer into independent monomers and thus facilitates substrate transport. Therefore, the phosphorylation-induced enhancement of substrate transport and dimer decoupling not only reconcile the competing experimental results but also provide an atomistic view of how nitrate transport is regulated in plants.

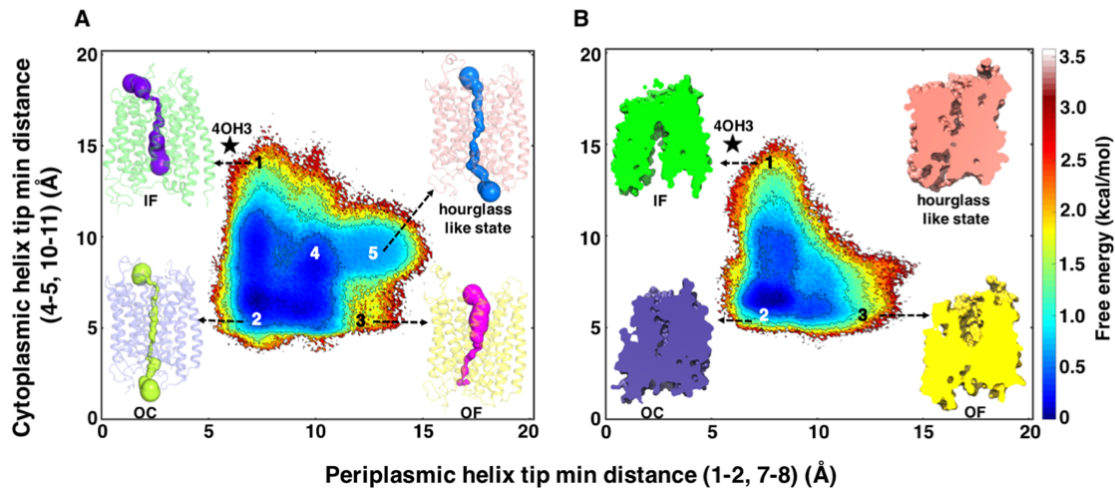


Figure 1. Conformational dynamics of (A) Unphosphorylated NRT1.1 and (B) phosphorylated NRT1.1 monomer. The opening and closure of channel gates are measured by the minimum helical tip distance of first four residues at the helices 1-2 and 7-8 at the periplasmic side and the helices 4-5 and 10-11 at the cytoplasmic side. The intermediate states are depicted as (1) IF, (2) OC, (3) OF, (4) partial IF-OF, and (5) hourglass-like state, respectively. The pore radius of the transporter channel for intermediate states was calculated using the HOLE program³⁸ and represented as a surface (Fig. 1A). The residues that drive the structural rearrangements and that stabilize the metastable conformational states are shown in ball and sticks (Fig. 1B). TPT shows two dominant pathways for UnpNRT1.1 from IF to OF via OC (1 → 2 → 3) and through partial IF-OF (1 → 4 → 3). For pNRT1.1, one dominant pathway was observed from IF to OF via OC (1 → 2 → 3).

- b. We have also discovered that the plant nitrate transporter is more efficient at transporting the nitrate fertilizers in its monomeric form as compared to its dimeric form. The difference in the free energy (blue color indicating higher stability) for the phosphorylated and unphosphorylated forms of the nitrate dimer shows that the phosphorylated dimer can undergo conformational transitions required for nitrate transport much faster than the unphosphorylated dimer. These observations have resolved a long standing question about how plants adapt to fluctuations in the soil nitrate concentration. Here, we have shown that the high affinity transport (under low nitrate availability) happens when transporter is in the monomeric phosphorylated form and the low affinity transport (under high nitrate availability) happens in the dimeric unphosphorylated form of the transporter.

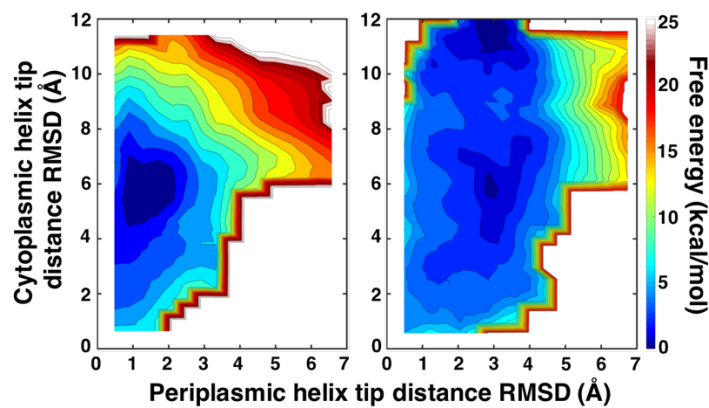


Figure 2. Conformational dynamics of UnpNRT1.1 and pNRT1.1 dimers. Free energy landscapes of (left) Unphosphorylated NRT1.1 and (right) phosphorylated NRT1.1 dimers are projected along the distance root mean square deviation (dRMSD) of the periplasmic and cytoplasmic helix tip in reference to the crystal structure (PDB: 4OH3).

Specific Aim 2: Validating conformational ensemble of transporters using DEER spectroscopy or other experimental data.

- a. Double electron–electron resonance (DEER) spectroscopy is a powerful experimental technique for understanding the conformational heterogeneity of proteins. It involves attaching nitroxide spin labels to two residues in the protein to obtain a distance distribution between them. However, the choice of residue pairs to label in the protein requires careful thought, as experimentalists must pick label positions from a large set of all possible residue-pair combinations in the protein. In this article, we address the problem of the choice of DEER spin-label positions in a protein. For this purpose, we utilize all-atom molecular dynamics simulations of protein dynamics, to rank the sets of labeled residue pairs in terms of their ability to capture the conformational dynamics of the protein. Our design methodology is based on the following two criteria: (1) An ideal set of DEER spin-label positions should capture the slowest conformational-change processes observed in the protein dynamics, and (2) any two sets of residue pairs should describe orthogonal conformational-change processes to maximize the overall information gain and reduce the number of labeled residue pairs. We utilize Markov state models of protein dynamics to identify slow dynamical processes and a genetic-algorithm-based approach to predict the optimal choices of residue pairs with limited computational time requirements. We predict the optimal residue pairs for DEER spectroscopy in β_2 adrenergic receptor, the C-terminal domain of calmodulin, and peptide transporter PepTSo. We find that our choices were ranked higher than those used to perform DEER experiments on the proteins investigated in this study. Hence, the predicted choices of DEER residue pairs determined by our method provide maximum insight into the conformational heterogeneity of the protein while using the minimum number of labeled residues.

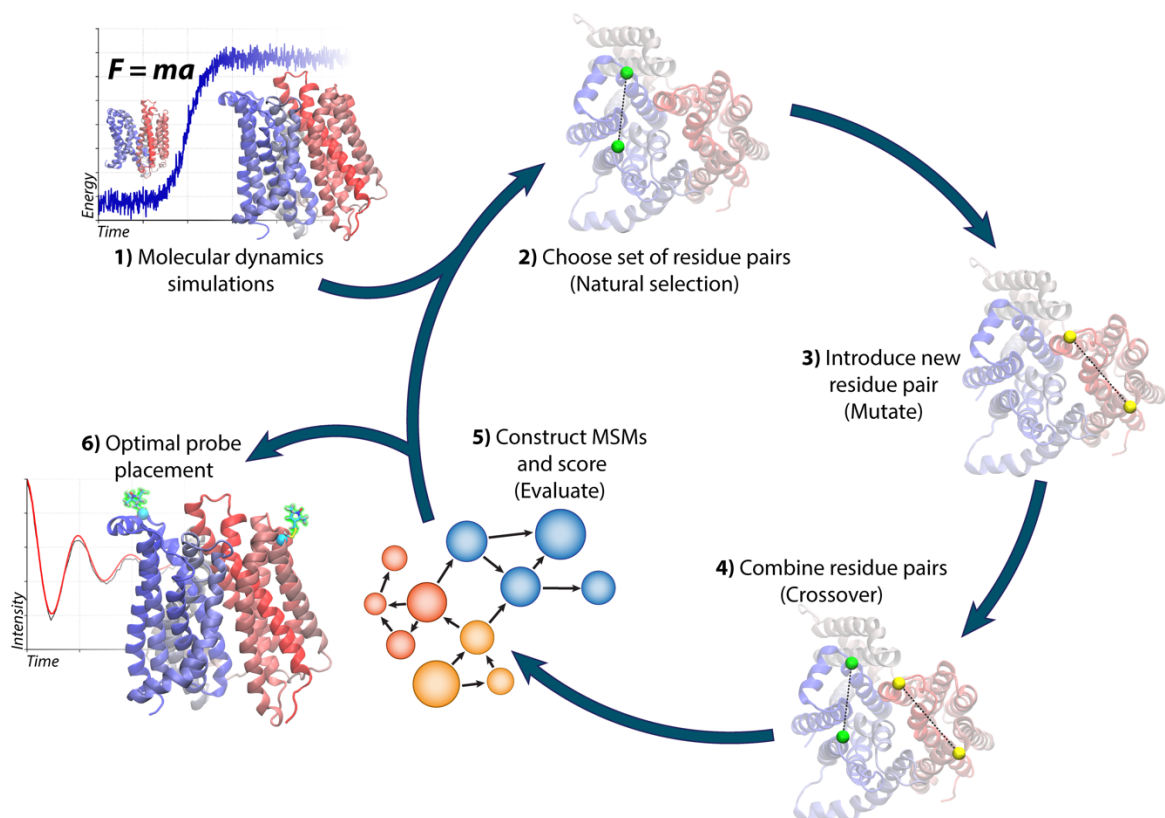


Figure 3. Markov state model (MSM)-guided experimental design for optimal placement of spin labelling probes. An advantage of the MSM is its ability to characterize long timescale behavior, such as conformational transitions observed in electron paramagnetic resonance spectroscopy. Shown here is a genetic algorithm implementation to utilize MSM to objectively identify key probe positions for distance measurement [25]. (1) MD simulations are first conducted to explore the conformational heterogeneity of the transporter. (2) A set of residue pairs is randomly chosen from all possible pair combinations. This step is analogous to natural selection in population genetics. Only residue pairs that can be experimentally measured will be considered. For example, residues buried in the transporter core or around the periphery will not be selected. (3) In some sets, a residue pair will be “mutated” in which that pair will be replaced with a new residue pair. (4) “Crossover” allows sets to combine residue pairs to further increase genetic diversity. (5) The sets of residue pairs are used to construct a MSM. Each MSM is quantitatively score with the generalized matrix Rayleigh quotient (GMRQ), which measures the MSM ability to capture long-timescales processes [14]. For the next iteration, the population of distances that obtained the highest GMRQ score “survives” as the next generation and progresses through the subsequent iteration of the genetic algorithm. (6) Upon convergence of the GMRQ score, the genetic algorithm stops and the distance pairs that obtained the highest GMRQ score is suggested for optimal probe placement.

b. In our previous work in part (a) done as part of this grant, we have proposed the use of

molecular dynamics simulation datasets to methodically pick residue pairs for DEER/EPR spectroscopy measurements. DEER/EPR measures the dipolar interaction between two paramagnetic loci in the protein and provides a distance distribution. Hence, it is a rich source for structural information of proteins. In the current work, we extend the earlier method for MD simulation guided prediction of residue pair choices for kinetics experiments. A rationally chosen set of residue pairs for biophysical experiments should, firstly be able to characterize the dynamics that occur at a slow timescale. Slow kinetics between conformations of the proteins indicates regions on the protein's energy landscape that are stable and require a high energy barrier to transition. These are often the pathways that are critical for the biological function of proteins, such as for a protein to fold into its native state. Secondly, the choice of residue pairs must be such it captures conformations that have not been observed by other residue pair measurements. In such a manner, we aim to maximize information about protein dynamics with minimum distance measurements. In the second year of the grant, we demonstrate the technique for three separate kinetics experiments, Trp-Tyr fluorescence quenching, luminescence resonance energy transfer (LRET) and triplet-triplet energy transfer (TTET). In the Figure 2, we show the prediction of probe positions for beat2-adrenergic receptor for the LRET technique. Here, we report GMRQ scores (indicating how good is the choice of experiments) for all possible LRET experiments that can be done for this protein.

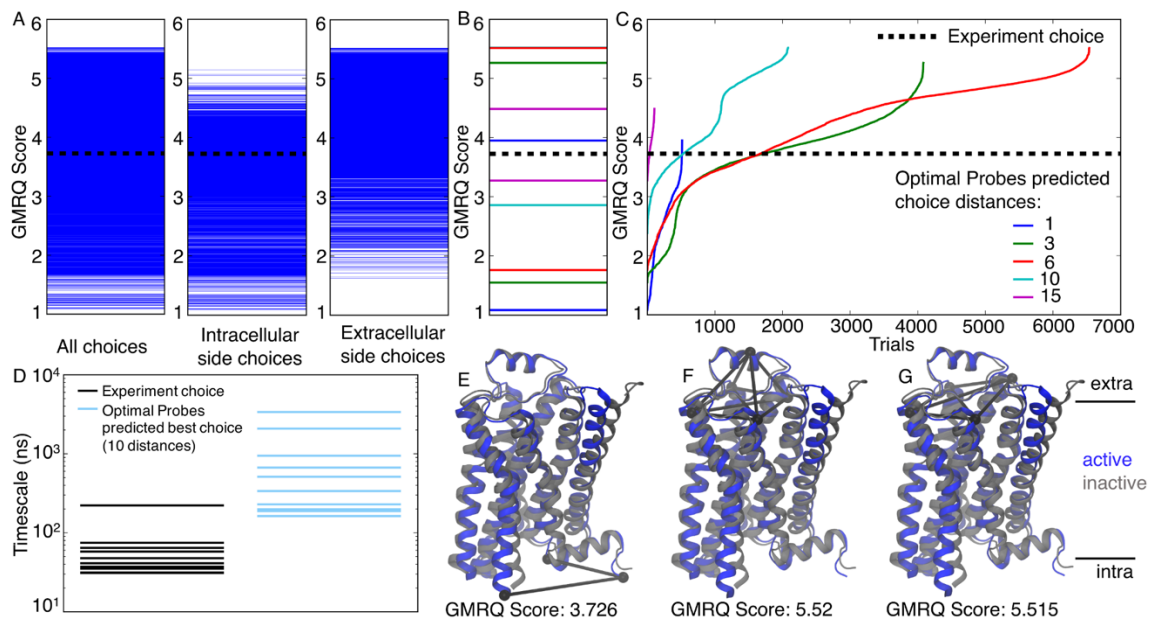


Figure 4. Optimal Probes predictions for LRET on β 2-Adrenergic receptor (β 2-AR). (A) GMRQ scores indicated the score received by all possible LRET experiments that can be done for this protein. The second and third columns indicate the scores for experiments that have only intracellular side distances and extracellular side distances, respectively. The score for the real LRET experiments done on this protein is shown in dashed black



line. (B) Best and lowest score with different number of residue pairs (or distances) in the trial set, 15 (magenta), 10 (cyan), 6 (red), 3 (green) or 1 (blue). (C) Scores for the trial sets of residue pairs differentiating the number of residue pairs (or distances) measured in the set. (D) Comparing top 10 implied timescales for the experimental distances MSM (black) and Optimal Probes best choice distances MSM (skyblue) shows that experiments already reported in the literature are not optimal (E) Cartoon representation of β 2AR showing the experimental choice residue pairs. (F) Cartoon representation of β 2AR showing the Optimal Probes predicted best set of residue pairs with 10 distances. (G) Cartoon representation of β 2AR showing the Optimal Probes predicted best set of residue pairs with 6 distances.

- c. Spectroscopy experiments are crucial to study membrane proteins for which traditional structure determination methods still prove challenging. Double electron-electron resonance (DEER) spectroscopy experiments provide protein residue-pair distance distributions that are indicative of their conformational heterogeneity. Atomistic molecular dynamics (MD) simulations are another tool that have been proven to be vital to study the structural dynamics of membrane proteins such as to identify inward-open, occluded, and outward-open conformations of transporter membrane proteins, among other partially open or closed states of the protein. Yet, studies have reported that there is no direct consensus between the distributional data from DEER experiments and MD simulations, which has challenged validation of structures obtained from long-timescale simulations and using simulations to design experiments. Current coping strategies for comparisons rely on heuristics, such as mapping the nearest matching peaks between two ensembles or biased simulations. Here we examine the differences in residue-pair distance distributions arising due to the choice of membranes around the protein and covalent modification of a pair of residues to nitroxide spin labels in DEER experiments. Through comparing MD simulations of two proteins, PepTSo and LeuT—both of which have been characterized using DEER experiments previously—we show that the proteins' dynamics are similar despite the choice of the detergent micelle as a membrane mimetic in DEER experiments. On the other hand, covalently modified residues show slight local differences in their dynamics and a huge divergence when the oxygen atom pair distances between spin labeled residues are measured rather than protein backbone distances. Given the computational expense associated with pairwise MTSSL labeled MD simulations, we examine the use of biased simulations to explore the conformational dynamics of the spin labels only to reveal that such simulations alter the underlying protein dynamics. Our study identifies the main cause for the mismatch between DEER experiments and MD simulations and will accelerate the development of potential mitigation strategies to improve the match.

Specific Aim 3: Identifying mechanisms responsible for enhanced nitrate uptake efficiency.

Major facilitator superfamily (MFS) transporter proteins operate via three different mechanisms: uniport, symport, and antiport. Despite extensive investigations, the molecular understanding of these transporters is less advanced than that of other transporters due to the complex coupling between substrates and the lack of distinct structures. We employ

extensive all-atom molecular dynamics simulations to dissect the complete substrate exchange cycle of the bacterial $\text{NO}_3^-/\text{NO}_2^-$ antiporter, NarK and $\text{Na}^+/\text{NO}_3^-$ symporter, NarU. We obtain the outward-open, occluded and inward-open state of these proteins and elucidate their substrate transport mechanism. Conformational transition occurs only in the presence of substrate, which weakens the electrostatic repulsion and stabilizes the transporter. Overall, this work presents a general working model for the antiport and symport mechanism for transport of nitrate ions.

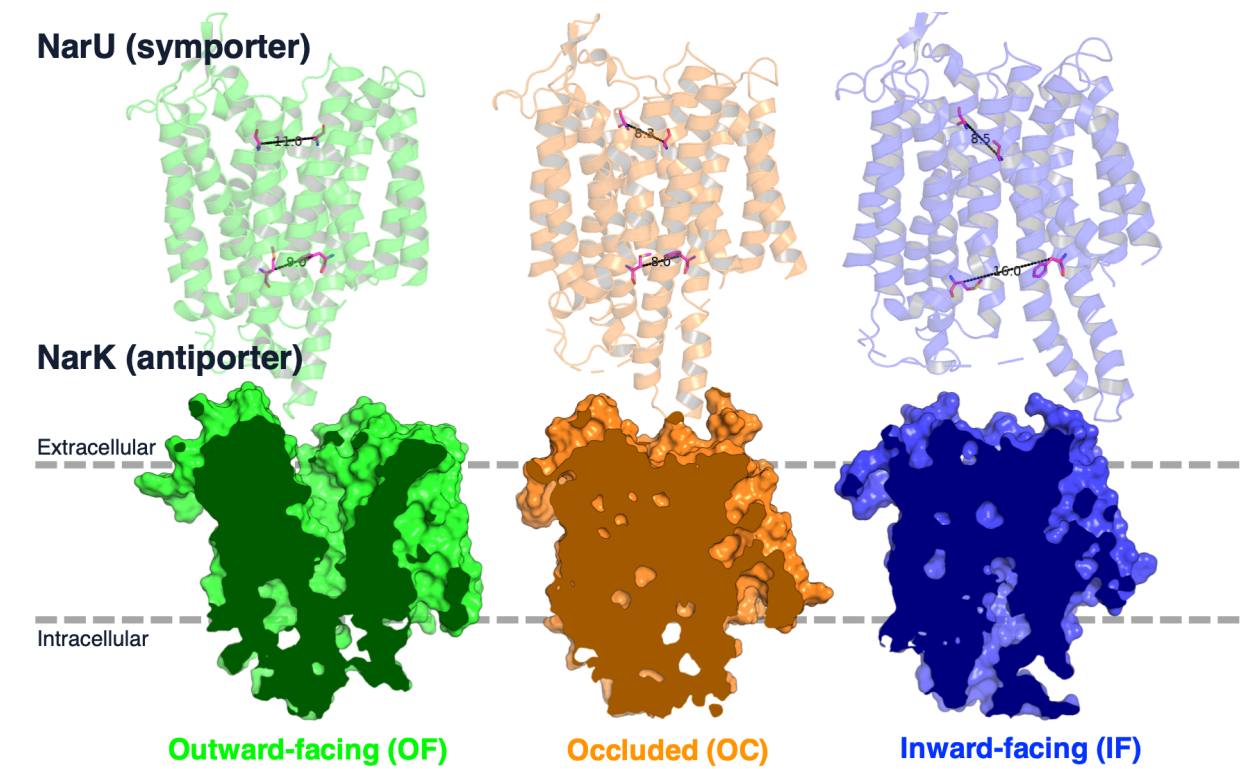


Figure 5. Predicted structures of bacterial NarU and NarK nitrate transporters. These transporters follow a similar rocker-switch type mechanism for substrate transport but they have different mechanisms. NarU acts a symporter i.e. it transports the sodium ion and nitrate from one side of the membrane to the other whereas NarK is an antiporter i.e. it transports the sodium ion from one side of the membrane and the nitrate from the opposite side of the membrane.

- 2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.
Include approaches taken to ensure robust and unbiased results.

We have used a variety of approaches to ensure robust and unbiased results.



1. **Using long time scale simulation data** for investigating the molecular mechanism of nitrate transport. One of the big limitations of simulations is the lack of sufficient simulation data. In these studies conducted as part of this study, we have used Blue Waters Supercomputer to generate extensive simulation data for the protein dynamics, which provides reliable and unbiased results.

2. **Using rigorous statistical tests to ensure the reliability of the models.** We have used the simulation data to build kinetic models of protein dynamics. We use Chapman-Kolmogorov test to make sure that the model are statistically sound and provide a reliable representation of the simulation data.

3. **Using multiple force fields to investigate the impact of systematic errors in the simulations.** One of the key systematic limitation of the molecular simulations is choice of appropriate force field or simulation parameters. We have performed simulations with multiple force fields to ensure that these different parameter sets provide similar results.

2.8. List key stakeholders that could be served by results of this project.

1. **Plant Biologist** working on improving nitrogen use efficiency of crops.
2. **Biophysicists** working on understanding the molecular mechanisms of substrate transport by membrane transporters.
3. **Plant breeders** interested in exploring avenues for plant transporter engineering for improve nitrogen-uptake efficiency.

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)



Students supported by this award participated in several educational outreach activities.

1. **The students helped PI in organizing summer camp (CURIE) for high school girls at University of Illinois** and taught them about protein structure and function using physical models of proteins made using soft foam. The camp also involved a hands-on activity in the form of a computer game that focused on how physical forces and interactions between atoms lead to large changes in proteins that enable them to function as molecular machines.
 2. **Students helped PI in organizing a one day visit by high school students for summer camps Discover Engineering and Evaluate Your option.** The camp activity was designed to teach students about how membrane transporters work in plants and human body to translocate molecules.
 3. **PI was invited by the American Society of Plant Biologists to prepare a teaching tool to introduce computational chemistry methods to senior undergraduate and graduate students in plant biology.** The teaching tool combines up-to- date peer-reviewed research-based content with flexible presentation components that can be used alone or integrated into lesson plans for presentation in a classroom. In particular, this teaching tool includes a short essay introducing each topic, PowerPoint slides, and suggested readings. The lecture notes are published as an article in the journal The Plant Cell. This teaching tools was recently highlighted on the American Society of Plant Biologists.
 4. **PI Shukla has given invited lectures in the Department of Plant Biology at University of Illinois and Cornell University.** These lectures are part of the courses on Computational Biology (Cornell University) and Biology in today's world (University of Illinois).
 5. Finally, we have also **written a comprehensive review on the role of molecular dynamics simulations in studying the post-translational modification of proteins.** Posttranslational modifications (PTMs) are an integral component to how cells respond to perturbation. While experimental advances have enabled improved PTM identification capabilities, the same throughput for characterizing how structural changes caused by PTMs equate to altered physiological function has not been maintained. In this Perspective, we cover the history of computational modeling and molecular dynamics simulations which have characterized the structural implications of PTMs. We distinguish results from different molecular dynamics studies based upon the timescales simulated and analysis approaches used for PTM characterization. Lastly, we offer insights into how opportunities for modern research efforts on in silico PTM characterization may proceed given current state-of-the-art computing capabilities and methodological advancements.
- 2.10. Describe challenges or delays encountered during project and actions that weretaken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

No challenges or delays encountered.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

The students supported from this grant were actively involved in educational outreach activities during the reporting period.

1. **Students participated in educational outreach activities.** The students helped PI in organizing summer camp CURIE (Catalyzing UR Interest in Engineering) for high school students at University of Illinois (July 2018-2021) and taught them about protein structure and function using physical models of proteins made using soft foam. The camp also involved a hands-on activity in the form of a computer game that focused on how physical forces and interactions between atoms lead to large changes in proteins that enable them to function as molecular machines.
2. **Students and PI had multiple opportunities to improve their presentation skills.**
 - a. Graduate student Jiangyan Feng presented her work in the American Chemical Society meeting and won the Chemical Computing Group Excellence Award for the graduate student from the Division of Computers in Chemistry Jiangyan Feng also received the Drickamer fellowship for outstanding research performance from the Chemical & Biomolecular Engineering department at University of Illinois.
 - b. Graduate student Shriyaa Mittal received the Dissertation completion fellowship from the Graduate College at University of Illinois. This fellowship is given to senior graduate students who have demonstrated outstanding research performance in the graduate school.
 - c. Graduate student Jiangyan Feng and Shriyaa Mittal presented their research in the Biophysical Society meeting in 2018-2020.
 - d. PI Shukla presented the results obtained in this grant period at the Annual Meeting of American Institute of Chemical Engineers in the sessions of Computational Molecular Science and Engineering Forum. PI also presented these results at the American Chemical Society meeting. PI gave multiple



invited talks at universities highlighting the research conducted as part of this grant.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

Undergraduate student-1; Graduate student-3; Postdoctoral Fellow-1.

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

The information products produced during this reporting period resulting directly from the FFAR award include (1) scholarly publications, reports or monographs or conference proceedings (2) models based on simulation data in the form of molecular dynamics trajectories which will be made available on request (3) software that enable experimental investigations of the proteins studied as part of this grant.

- 3.2. Please provide a list of citations for the information products produced during the project.

Conference Presentations:

1. D. Shukla, High Throughput Experiments and Simulations for Machine Learning Guided Protein Engineering. Department of Chemical Engineering, University of New Hampshire, Mar **2022**.
2. D. Shukla, Computational Structural Biology of Plants. School of Integrative Plant Science Plant Biology Section, Cornell University, Mar **2021**.
3. D. Shukla, Computational Structural Biology of Plants. Department of Chemical & Biological Engineering, Cornell University, Mar **2021**.
4. D. Shukla, Optimal Design of Spectroscopy Experiments for Probing Protein Dynamics. PHYS Division Award Symposium. American Chemical Society Fall meeting, Aug **2021**.
5. D. Shukla, High Throughput Experiments and Simulations for Machine Learning Guided Protein Engineering., Area 15D/E session on Design, Engineering, and Structural Prediction of Peptides and Proteins, AIChE Annual Meeting Oct **2021**.
6. D. Shukla, Elucidating mechanisms of substrate transport via membrane transporters. Department of Chemical Engineering, North Carolina State University, Sep **2020**.

7. D. Shukla Elucidating mechanisms of substrate transport via membrane transporters. Department of Bioengineering, University of Illinois at Urbana-Champaign, Sep **2020**.
8. S. Mittal, D. Guin, M. Gruebele and D. Shukla. Is dodine a protein stabilizer or destabilizer? It's complicated. Biophysical Society Annual Meeting, San Diego, CA, 2020.
9. S. Mittal and D. Shukla. Reconciling membrane protein simulations with experimental spectroscopic data. Biophysical Society Annual Meeting, San Diego, CA, 2020.
10. J. Feng and Shukla D. Elucidating molecular mechanism of nitrate transporters. Global Change and Photosynthesis Research Unit (GCPRU) Seminar Series, Department of Crop Sciences, University of Illinois at Urbana-Champaign, 2019.
11. J. Feng and Shukla D. How Nark antiports nitrate and nitrite? Mechanisms of Membrane Transport, Gordon Research Conference, Colby-Sawyer College, New London, 2019.
12. C. Zhao, J. Chen, F. Aldukhi, A. S. Moffett and Shukla D. Quantitative characterization of protein-ligand and protein-protein binding processes involved in plant hormone signaling. AIChE Annual Meeting, Orlando, FL, 2019.
13. C. Zhao and D. Shukla Towards rational design of chemical modulators to improve plant drought tolerance. AIChE Annual Meeting, Orlando, FL, 2019.
14. C. Zhao and D. Shukla. Molecular basis and engineering of plant drought resistance signaling. Great Lakes Chinese American Chemical Society 23rd Annual Conference, Chicago, IL, 2019.
15. S. Mittal and D. Shukla. Simulation Guided Design of Spectroscopy Experiments Via Maximizing Kinetic Information Gain. Biophysical Society Meeting, Baltimore, 2019.
16. J. Feng and D. Shukla. Reinforcement Learning of Protein Conformational Ensemble. Biophysical Society Meeting, Baltimore, 2019.
17. D. Shukla. Elucidating Mechanisms of Substrate Transport in Membrane Transporters. Biophysical Society Meeting, Baltimore, 2019.
18. C. Zhao and D. Shukla. Complex dynamics of hormone perception and receptor activation in plant drought resistance signaling. Gordon Research Conference on Plant Molecular Biology, Holderness, NH, 2018.
19. D. Shukla. Methodologies for enhanced unbiased sampling of the free energy landscapes of proteins. American Institute of Chemical Engineers Annual Meeting, Pittsburgh, 2018.
20. D. Shukla. Enhanced unbiased sampling of the conformational free energy landscapes of proteins. American Chemical Society Fall Meeting, Boston, 2018.
21. Chuankai Zhao and D. Shukla. Complex dynamics of hormone perception and receptor activation in plant drought resistance signaling. American Chemical Society Fall Meeting, Boston, 2018.

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Yes, we submitted four manuscripts directly related to the research conducted during this reporting period and they have already been accepted for publication.

Scholarly Publication(s): **denotes corresponding author and underline denotes students supported by FFAR*

1. *S. Mittal, S. Dutta and **D. Shukla**, Reconciling Membrane Protein Simulations with Experimental DEER Spectroscopy Data. *Phys. Chem. Chem. Phys.*, Vol. 25, pages 6253-6262. doi: 10.1039/D2CP02890E.
2. *C. Zhao and **D. Shukla**, Molecular basis of the activation and dissociation of dimeric PYL2 receptor in abscisic acid signaling. *Phys. Chem. Chem. Phys.*, Vol. 24, pages 724-734, **2022**. doi: 10.1039/D1CP03307G
3. *J. Feng, B. Selvam, and **D. Shukla**, How antiporters exchange substrates across the cell membrane? An atomic-level description of the complete exchange cycle in NarK. *Structure*, Vol. 29(8), pages 922-933, **2021**. doi: 10.1016/j.str.2021.03.014
4. *A. T Weigle, J. Feng and **D. Shukla**, Thirty years of molecular dynamics simulations on post-translational modifications of proteins, *Physical Chemistry Chemical Physics*, Volume 24, Pages 26371-26397, **2022**. doi: 10.1039/d2cp02883b
5. *J. Feng and **D. Shukla**, Fingerprint Contacts: Predicting Protein Alternative Conformations from Coevolution. *J. Phys. Chem. B*, Vol. 124(18), pages 3605-3615, **2020**. doi: 10.1021/acs.jpcc.9b11869
6. *B. Selvam, J. Feng and **D. Shukla**, Atomistic Insights Into The Mechanism of Dual Affinity Switching In Plant Nitrate Transporter NRT1.1. *bioRxiv*, **2022**. doi: 10.1101/2022.10.17.512638.
7. *J. Feng, J. Chen, B. Selvam and **D. Shukla**, Computational Microscope: Unravelling the Hidden Life of Plants, *The Plant Cell*, 2019. doi: doi:10.1105/tpc.119.tt121
8. *D. Guin, S. Mittal, B. Bozymski, **D. Shukla*** and M. Gruebele*, Dodine acts as a Kosmo-Chaotropic Agent. *J. Phys. Chem. Lett.*, Vol. 10(10), 2600-2605, 2019.
9. *J. Park, B. Selvam, **D. Shukla*** and E. Procko*. Selection of the Dimeric Sweet Taste Receptor for Surface Expression and Inter-subunit Interactions. *J. Biol. Chem.*, Vol. 294, 4759-4774, 2019.
10. *B. Selvam, S. Mittal and **D. Shukla**, Free energy landscape of the complete transport cycle in a key bacterial transporter. *ACS Central Science*, 2018. Doi: 10.1021/acscentsci.8b00330.
11. *S. Mittal and **D. Shukla**, Maximizing Kinetic Information Gain of Markov State Models for Optimal Design of Spectroscopy Experiments. *The Journal of Physical Chemistry B*, 2018. Doi: 10.1021/acs.jpcc.8b07076.
12. *S. Mittal and **D. Shukla**, Recruiting Machine Learning Methods for Molecular Simulations of Proteins. *Molecular Simulation*, 2018. Doi: 10.1080/08927022.2018.1448976.

13. *C. Zhao and **D. Shukla**, SAXS-guided Enhanced Unbiased Sampling for Structure Determination of Proteins and Complexes. Scientific Reports, Vol. 8, Article number: 17748, 2018.

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

1. We have provided the materials related teaching tools in plant biology on box.illinois.edu and plantae.org. Link: <https://community.plantae.org/path/5375167063436297710>
2. We have provided the structures of the peptide transporter for public use on Zenodo. These structures were published recently as part of the manuscript resulting directly from the FFAR grant. Link: <https://zenodo.org/record/1402798>
3. We have posted the Optimal probes software developed as part of this award on the Github for public use: Link: <https://github.com/ShuklaGroup/optimalProbes>
4. We have posted the SAXS based conformational sampling software developed as part of this award on the Github for public use: <https://github.com/ShuklaGroup/SAXS-guidedAdaptiveSampling>
5. We have posted the Fingerprint contact software developed as part of this award on the Github for public use. Fingerprint contact: <https://github.com/ShuklaGroup/FingerprintContacts>.
6. A total of 300 μ s simulation data were obtained and used for further data analysis. The in-house python codes used to perform simulations and analysis scripts are available at https://github.com/ShuklaGroup/NarK_Structure_2021_Files.
- a. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

Nothing to Report

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
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Patents	#	#	
Copyrights	#	#	
Trademarks	#	#	
Inventions	#	#	

- b. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

Nothing to Report

- c. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?

We have adopted following strategies to ensure that others can easily discover and access the listed information products. (i) We have published the FFAR supported research manuscripts in open access journal such as our recent work on membrane transporters in ACS central science. (ii) If the research manuscripts are published in traditional journals with limited or expensive open-access publishing option, we put the open-access version of the manuscripts on arXiv, biorXiv or chemRxiv open-access repositories. (iii) The datasets generated as part of this grant will be made publicly available on open-access data repositories such as figshare. We will also provide links to these repositories on our group website and in the open-access versions of the published manuscripts.

7. Data Management



- a. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

Yes, project has generated several datasets related to molecular dynamics simulations of membrane transporters involved in plant nitrate transport. These datasets contain simulation trajectories that describe the position of all atoms in the system as a function of time. We have collected the following datasets: (i) simulation dataset for Pepto (size 2.1 TB), (ii) simulation datasets for plant nitrate transporter NRT1.1 (7.0 TB) and (iii) simulation data for bacterial nitrate transporter, NarK (3.5 TB) and (iv) initial simulations of the bacterial nitrate transporter NarU (3 TB)

- b. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

These datasets all describe the mechanism of nitrate transport via membrane transporters in different organisms and these datasets are generated using the same simulation techniques on Blue Waters Supercomputer.

- c. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

Given the size of the datasets, it is not possible to share these datasets in a compressed folder. Current practice in the field is to make the data available to the scientific community on request. We have now uploaded these datasets to box.illinois.edu and the link to the box folder can be shared with the researchers immediately on request.



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RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: _____ Diwakar Shukla _____

PI Signature: Diwakar Shukla Date: _____ 04/06/2023 _____

Authorized Signing Official (ASO) Name: Paul N. Ellinger

ASO Signature: _____ Date: 04/12/2023 _____